

Cambridge IGCSE™

BIOLOGY**0610/41**

Paper 4 Theory (Extended)

October/November 2025**MARK SCHEME**Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **14** printed pages.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1 Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.

2 The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.

3 Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).

4 The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.

5 'List rule' guidance

For questions that require ***n*** responses (e.g. State **two** reasons ...):

- The response should be read as continuous prose, even when numbered answer spaces are provided.
- Any response marked *ignore* in the mark scheme should not count towards ***n***.
- Incorrect responses should not be awarded credit but will still count towards ***n***.
- Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response.
- Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	point has been noted, but no credit has been given or blank page seen
	correct awarding one mark from marking point or marking group 1. similar numbered ticks are used for marking point or marking groups 2, 3, 4 etc.

Annotation	Meaning
	pages are linked together
	used to highlight part of the response
	used to highlight parts of an extended response
	used to highlight parts of an extended response
	Point already given
	Maximum mark reached
	Key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	Maximum number of marks for a marking point has been awarded.

Mark Scheme Abbreviations:	
;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
<u>underline</u>	actual word given must be used by candidate (grammatical variants excepted)
()	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
MP	marking point

Question	Answer	Marks	Guidance
1(a)	go to 2		3 6 correct = 3 3 or 4 or 5 correct = 2 1 or 2 correct = 1
	go to 4		
	go to 3		
	<i>Vibrio cholerae</i>	D	
	<i>Helicobacter pylori</i>	B	
	<i>Salmonella typhi</i>	E	
	<i>Treponema pallidum</i>	C	
	go to 5		
	go to 6		
	<i>Staphylococcus aureus</i>	A	
	<i>Streptobacillus moniliformis</i>	G	
	<i>Streptococcus pyogenes</i>	F	
	1(b)(i)	<i>radius:</i> 0.25 um ; <i>volume:</i> 0.39 um ³ ;;;	
1(b)(ii)	prokaryote ;	1	
1(c)(i)	J protein coat ; K genetic material ;	2	

Question	Answer	Marks	Guidance
1(c)(ii)	<i>any two from:</i> cell wall ; cell membrane ; circular / loop of, DNA ; plasmid ; cytoplasm ; ribosome ; AVP ;;	2	e.g. flagella

Question	Answer			Marks	Guidance
2(a)(i)	letter on Fig. 6.1	name of the blood vessel	carries oxygenated blood	4	one mark for each correct row
	A	pulmonary artery	no		
	B	aorta	yes		
	C	hepatic portal vein	no		
	D	renal vein	no		
2(a)(ii)	name: valve ; function: prevents back flow of blood / AW ;			2	
2(b)(i)	any two from: (only) two chambers ; no, septum / left and right sides / AW ; fewer valves ; heart (chamber) walls of similar thickness (to each other) ; fewer blood vessels (going to and from the heart) ;			2	ora throughout

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Question	Answer	Marks	Guidance
2(b)(ii)	<i>any four from:</i> 1 blood is pumped to the lungs then to the body / blood goes through the heart twice in one complete journey / double circulation ; 2 oxygenated and deoxygenated blood, are kept separate / do not mix / separated by septum ; 3 efficient, supply of blood / glucose / oxygen / nutrients (to, the body / AW) ; 4 efficient, removal / carbon dioxide / urea / lactic acid (from, body / AW); 5 provides high(er) pressure (in body) ; 6 which allow a high, metabolic <u>rate</u> / <u>rate</u> of respiration ; 7 which allows filtration in kidneys (for excretion) ; 8 low(er) pressure in, pulmonary, artery / circuit / AW ; 9 to prevent damage to (capillaries in the) lungs ; 10 allow more time for, gas exchange / absorption of oxygen ; 11 AVP ;	4	e.g. larger / steeper, diffusion gradient between capillaries and respiring tissues / allows large body size / maintenance of body temperature
2(c)(i)	red blood cell ;	1	
2(c)(ii)	actual size = image size ÷ magnification ;	1	
2(c)(iii)	31.5 (µm) ;	1	

Question	Answer	Marks	Guidance
3(a)(i)	A denitrification ; B nitrification ;	2	
3(a)(ii)	<i>any two from:</i> lightning ; bacteria ; AVP ;	2	e.g. Haber process
3(a)(iii)	ammonia / ammonium (ions) ;	1	
3(a)(iv)	liver ;	1	
3(a)(v)	removal of nitrogen containing part (of amino acids / protein) ; urea is formed ;	2	
3(b)	greenhouse effect ; rice ; carbon dioxide ; photosynthesis ; burning / combustion ;	5	
3(c)	<i>any six from:</i> 1 untreated sewage / fertiliser / nutrients ; 2 increased availability of nitrates / ions / AW ; 3 causing algal bloom / algae reproduction ; 4 algae reduce (penetration of) sunlight (in the water) / AW ; 5 (so rooted) plants unable to photosynthesise ; 6 so plants die ; 7 bacteria, decompose / feed, on dead, plants or algae ; 8 so bacterial populations increase ; 9 bacteria respire aerobically ; 10 bacteria use up the oxygen in the water ; 11 organisms / fish , die / suffocate / migrate, due to lack of oxygen ;	6	A decomposers throughout for bacteria

Question	Answer	Marks	Guidance
4(a)(i)	<i>any three from:</i> scent ; nectar ; large / colourful, petals ; pollen (as a source of food) ; AVP ;	3	e.g. honey guides / ref. to landing platforms / ref. to mimicking insects
4(a)(ii)	<i>any one from:</i> large ; sticky / spiky / AW ;	1	
4(b)	<i>any two from:</i> (<i>monocots have</i>) narrow / strap / long/AW, leaves ; parallel (leaf) veins ; scattered vascular, tissue / bundles, (in the stem) ; hollow stems ; petals in multiples of three ; fibrous roots ; one cotyledon ; pollen grains have one pore or furrow ; AVP ;;	2	
4(c)	filament labelled ; sepal labelled ; an ovule labelled ;	3	

Question	Answer	Marks	Guidance
4(d)	<i>any five from:</i> 1 pollen (released) from the anther ; 2 pollen transferred to stigma ; 3 <u>pollen tube</u> , grows ; 4 (pollen tube goes) down / through, the style ; 5 ref. to ovule / ovary ; 6 pollen / male, nucleus travels down the (pollen) tube / style ; 7 pollen / male, and, ovule / female, <u>nuclei</u> fuse ; 8 fertilisation ; 9 AVP ;	5	e.g. ref. to haploid nuclei or diploid zygote ;
4(e)(i)	sperm (cell) ;	1	
4(e)(ii)	testis / testes ;	1	
4(e)(iii)	meiosis ;	1	A reduction division
4(e)(iv)	make fluid / semen, for the sperm / male gametes, to swim / AW ;	1	A ref. to providing nutrients

Question	Answer	Marks	Guidance
5(a)(i)	transpiration / diffusion ;	1	
5(a)(ii)	<i>any five from:</i> <i>comparison</i> 1 both lose mass ; 2 higher humidity, less water vapour is lost ; 3 ref. to steeper rate / greater water lost, initially (for both groups) ; 4 comparative data quote with units ; <i>explanation for high humidity</i> 5 (higher humidity) greater amount of water vapour in the atmosphere ; 6 reduced water potential, gradient ; 7 less water vapour lost, through stomata / between guard cells ;	5	ora throughout

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Question	Answer	Marks	Guidance
5(a)(iii)	line drawn on the graph above the low humidity line ;	1	
5(b)(i)	<i>any three from:</i> glands in the skin produce sweat ; sweat on (the surface of) the skin evaporates ; using / AW, heat / energy (from the body) ; (internal) body temperature, decreases / returns, to a set point ; ref. to homeostasis / negative feedback ;	3	
5(b)(ii)	expiration / AW ;	1	

Question	Answer	Marks	Guidance
6(a)	nucleus ;	1	
6(b)	<i>any one from:</i> enzymes ; membrane carriers ; receptors for neurotransmitters ; AVP ;	1	e.g. insulin / glucagon / antibodies / keratin / haemoglobin
6(c)	<i>any three from:</i> (mRNA) is a copy of the gene / DNA / base sequence ; gene / DNA, remains in the nucleus ; (mRNA takes instructions) to cytoplasm / ribosome ; mRNA, passes through / attaches to / 'read by', ribosome ; base sequence determines sequence of amino acids (in the protein) ;	3	